

Shrushti Zod

Deploying a Model in the Browser

TRAIN

SAVE

RELOAD

PREDICT

EXPORT

IMPORT

01 TRAIN MODEL & SAVE TO LOCALSTORAGE

Trains a small dense neural network on the XOR problem (4 samples, 2 inputs → 1 output) and saves it to browser LocalStorage.

EPOCHS

300

LEARNED RATE

0.1

▶ TRAIN & SAVE TO LOCALSTORAGE

[01:04:19] Building model...

[01:04:19] Training for 300 epochs. lr=0.1.

[01:04:19] Epoch 00 - loss: 0.00010 acc: 1.000

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02 RELOAD FROM LOCALSTORAGE & VERIFY PREDICTIONS

Loads the saved model back from LocalStorage and runs predictions on all XOR inputs to verify they match the original model output.

↻ RELOAD & VERIFY PREDICTIONS

[01:04:02] Loading model from LocalStorage ("xor-model").

MODEL SUMMARY

Dense

8 units - relu

20

Dense

8 units - relu

70

Dense

1 units - sigmoid

50

Total Parameters: 508

TRAINING LOSS

STORAGE STATUS

LOCALSTORAGE:

Reload

EXPORTED FILES:

Exported

IMPORTED MODEL:

Loaded

<!DOCTYPE html>

<html lang="en">

<head>

</head>

<body>

</body>

</html>

body

element.style {

}

body {

background: var(--bg);

color: var(--text);

font-family: var(--sans);

min-height: 100vh;

background-image: linear-gradient(rgba(79, 142, 247, 0.03) 1px, transparent 1px), linear-gradient(90deg, rgba(79, 142, 247, 0.03) 1px, transparent 1px);

background-size: 40px 40px;

}

*, ::before, ::after {

box-sizing: border-box;

margin: 0;

padding: 0;

}

↻ RELOAD & VERIFY PREDICTIONS

[01:04:02] Loading model from LocalStorage ("xor-model").

[01:04:02] Model loaded ✓

[01:04:02] Predictions from reloaded model computed ✓

[01:04:02] All predictions match ✓

PREDICTION COMPARISON

INPUT	ORIGINAL	RELOADED	MATCH
[0,0]	0.000070	0.000070	✓ Yes
[0,1]	1.000000	1.000000	✓ Yes
[1,0]	0.000003	0.000003	✓ Yes
[1,1]	0.000000	0.000000	✓ Yes

03 EXPORT TO FILES & RE-IMPORT

Downloads the model as .json + .bin files, then re-imports them via file picker to run predictions again.

+ EXPORT MODEL FILES

+ IMPORT MODEL FILES

[01:04:04] Importing xor-model.weights.bin, xor-model.json.

[01:04:04] Model imported from files ✓

[01:04:04] Predictions from imported model computed ✓

[01:04:04] Import + prediction test complete ✓

POST-IMPORT PREDICTIONS

INPUT	EXPECTED (XOR)	IMPORTED MODEL	A CROWN
[0,0]	0	0.000070	0.00000
[0,1]	1	1.000000	0.00000
[1,0]	1	0.000003	0.00001
[1,1]	0	0.000000	0.00000

EXPORTED FILES:

Exported

IMPORTED MODEL:

Loaded

LAB OBJECTIVES

01. Train a small model and save it to LocalStorage

02. Reload the model and verify predictions match the original

03. Export model to files and re-import test predictions again

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